#9.

Eg. 9.15.

$$f(y,p) = f(y|p)\cdot f(p) = \left\{\binom{n}{y} p^{y}(1-p)^{n-y}\right\}\left\{\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\alpha-1}(1-p)^{\beta-1}\right\} = \binom{n}{y}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{y+\alpha-1}(1-p)^{n-y+\beta-1}$$

$$f(y) = \int_0^1 f(y,p)\,dp = \int_0^1 \binom{n}{y}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{y+\alpha-1}(1-p)^{n-y+\beta-1}\,dp$$

$$= \binom{n}{y}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 \underbrace{p^{y+\alpha-1}(1-p)^{n-y+\beta-1}}_{\text{// Def 2.28}}\,dp$$

$$B(\alpha',\beta')\frac{1}{B(\alpha',\beta')} p^{\alpha'-1}(1-p)^{\beta'-1}$$

$\alpha' = y+\alpha$

$\beta' = n-y+\beta$

$\alpha'+\beta' = n+\alpha+\beta$

$$= \binom{n}{y}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\cdot B(\alpha',\beta')\cdot \underbrace{\int_0^1 \text{Beta}(\alpha',\beta')\,dp}_{=1}$$

↓ Thm 2.27

$$= \binom{n}{y}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\cdot\frac{\Gamma(\alpha')\Gamma(\beta')}{\Gamma(\alpha'+\beta')}$$

$$= \binom{n}{y}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\cdot\frac{\Gamma(y+\alpha)\Gamma(n-y+\beta)}{\Gamma(n+\alpha+\beta)}$$

sample mean

prior mean.

$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$

$E[p]$

$$\hat{p}_B = \left(\frac{n}{\alpha+\beta+n}\right)\left(\frac{y}{n}\right) + \left(\frac{\alpha+\beta}{\alpha+\beta+n}\right)\cdot\left(\frac{\alpha}{\alpha+\beta}\right)$$

Eg. 9.1b.

$X_i|\theta \overset{iid}{\sim} N(\theta, \sigma^2)$

$\theta \sim N(\mu, \tau^2)$

$\Rightarrow$

$$f(x_1 \dots x_n|\theta) = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2}\left(\frac{x_i-\theta}{\sigma}\right)^2\right\}\right)^n = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left\{-\frac{1}{2}\sum_{i=1}^n\left(\frac{x_i-\theta}{\sigma}\right)^2\right\} = \underset{\text{constant}}{C} \exp\left\{-\frac{1}{2}\sum_{i=1}^n\left(\frac{x_i-\theta}{\sigma}\right)^2\right\}$$

$$f(\theta) = \frac{1}{\sqrt{2\pi\tau^2}} \exp\left\{-\frac{1}{2}\left(\frac{\theta-\mu}{\tau}\right)^2\right\} = \underset{\text{constant}}{D} \exp\left\{-\frac{1}{2}\left(\frac{\theta-\mu}{\tau}\right)^2\right\}$$

$\exp\{-a(\theta-b)^2\}$ $(a>0)$

正規分布の核

① $f(x_1 \dots x_n, \theta) = f(x_1 \dots x_n|\theta) \cdot f(\theta)$

$$= C \cdot D \cdot \exp\left\{-\frac{1}{2}\left[\underline{\sum_{i=1}^n\left(\frac{x_i-\theta}{\sigma}\right)^2 + \left(\frac{\theta-\mu}{\tau}\right)^2}\right]\right\}$$

$$\sum_{i=1}^n\left(\frac{x_i^2-2x_i\theta+\theta^2}{\sigma^2}\right) + \frac{\theta^2-2\theta\mu+\mu^2}{\tau^2}$$

$$= \frac{\sum_{i=1}^n x_i^2 - 2\theta\sum_{i=1}^n x_i + n\theta^2}{\sigma^2} + \frac{\theta^2-2\theta\mu+\mu^2}{\tau^2}$$

$$= -\frac{2\theta\sum_{i=1}^n x_i}{\sigma^2} + \frac{n\theta^2}{\sigma^2} + \frac{\theta^2}{\tau^2} - \frac{2\theta\mu}{\tau^2} + \underset{\text{constant}}{E}$$

$$= \theta^2\left(\frac{n}{\sigma^2}+\frac{1}{\tau^2}\right) - 2\theta\left(\frac{\sum_{i=1}^n x_i}{\sigma^2} + \frac{\mu}{\tau^2}\right) + E$$

$$= \theta^2 A - 2\theta B + E$$

$$A = \left(\frac{n}{\sigma^2}+\frac{1}{\tau^2}\right)$$

$$B = \left(\frac{\sum_{i=1}^n x_i}{\sigma^2} + \frac{\mu}{\tau^2}\right)$$

$$= A\left(\theta^2 - \frac{2\theta B}{A} + \frac{B^2}{A^2}\right) - \frac{B^2}{A} + E$$

$$= A\left(\theta - \frac{B}{A}\right)^2 - E'$$

$$= C \cdot D \cdot \exp\left\{-\frac{1}{2}\left[A\left(\theta-\frac{B}{A}\right)^2 - E'\right]\right\}$$

$$= C \cdot D \cdot F \cdot \exp\left\{-\frac{A}{2}\left(\theta-\frac{B}{A}\right)^2\right\}$$

$$= C \cdot D \cdot F \cdot \exp\left\{-\frac{1}{2}\frac{\left(\theta-\underset{\text{mean}}{\frac{B}{A}}\right)^2}{\underset{\text{Var}}{\frac{1}{A}}}\right\}$$

② $$f(x_1 \dots x_n) = \int_{-\infty}^{\infty} C \cdot D \cdot F \cdot \exp\left\{-\frac{1}{2}\frac{\left(\theta-\frac{B}{A}\right)^2}{\frac{1}{A}}\right\} d\theta = \text{constant}$$

③ $$f(\theta|x_1 \dots x_n) = \frac{f(x_1 \dots x_n, \theta)}{f(x_1 \dots x_n)} \propto \exp\left\{-\frac{1}{2}\frac{\left(\theta-\frac{B}{A}\right)^2}{\frac{1}{A}}\right\}$$

$$E[\theta] = \frac{B}{A} = \frac{\frac{\sum_{i=1}^n x_i}{\sigma^2} + \frac{\mu}{\tau^2}}{\frac{n}{\sigma^2}+\frac{1}{\tau^2}} = \frac{\frac{n\bar{x}\tau^2+\mu\sigma^2}{\sigma^2\tau^2}}{\frac{n\tau^2+\sigma^2}{\sigma^2\tau^2}} = \frac{n\bar{x}\tau^2+\mu\sigma^2}{n\tau^2+\sigma^2} = \frac{n\tau^2}{n\tau^2+\sigma^2}\bar{x} + \frac{\sigma^2}{n\tau^2+\sigma^2}\mu$$

$$Var[\theta] = \frac{1}{A} = \frac{\tau^2\sigma^2}{n\tau^2+\sigma^2}$$